

Dynamic interplay between preferential and matrix flows enhances water circulation in rubber agroforestry soil profiles

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ABSTRACT

Spatial variations in preferential and matrix flow play a critical role in the key processes related to ecohydrology, soil and water conservation, and groundwater pollution. However, the interactions between preferential and matrix flow in regulating water and solute transport through soil profiles remain inadequately understood, especially under the alternating rainy and dry seasons of tropical regions. In this study, we investigated how alternating mechanisms of preferential and matrix flow influence soil water storage in southwestern China. We used ecological bistable theory and in-situ field measurements across rubber-related forest systems with different root biomass (increasing in the following order: rubber monoculture (RM) < rubber-*Alpinia oxyphylla* (RAO) < rubber-orange-tea (ROT) < artificial rubber rainforest (ARR)). The results show that the soil water storage in RM to RAO, ROT, and ARR increased by 18 %, 23 %, and 31 %, respectively. These increases were strongly associated with soil total porosity ($R^2 = 0.933$), saturated hydraulic conductivity (K_s) ($R^2 = 0.879$), and water flow behavior ($R^2 = 0.968$). The dominant water flow behaviors during the alternating rainy and dry seasons were high-low levels of matrix flow and low level of preferential flow in RM, and high-low levels of preferential flow and high level of matrix flow in ROT and in ARR. An opposite relationship occurred between preferential and matrix flows, with matrix flow exhibiting a sinusoidal pattern at the annual scale. Preferential flow dominated downward water flow in the rainy season (contributing ~13 %, 58 %, 61 %, and 72 % in RA, RAO, ROT, and ARR), whereas matrix flow dominated upward water flow in the dry season (accounting for 33 %, 62 %, 73 %, and 76 %, respectively). The interaction between preferential and matrix flows formed a bidirectional water movement throughout the soil profiles, enhancing water transport and storage across the forest systems, which increased in the order of RM < RAO < ROT < ARR. This study elucidates the bistable dynamics between preferential and matrix flows in rubber-related forests, offering insights for balancing rainwater utilization and intercropping design in rubber-based agroforestry management.

1. Introduction

Agroforestry systems, which integrate multiple species and vertical structures, provide advantages such as high productivity, valuable profits, sustainable land use, and optimal utilization of soil and water resource (Pavlidis and Tsihrantzis, 2017). The promotion of agroforestry systems is increasingly urgent under the projected intensification of

climate change and drought (IPCC, 2021). This is the case in Xishuangbanna, a humid tropical region of Southeast Asia, where rapid rubber plantation expansion has induced severe hydrological and geological issues (Li et al., 2012; Liu et al., 2014). In response, rubber-based agroforestry systems have been established to mitigate these detrimental effects and have shown potential in restoring degraded ecosystem services (van Noordwijk et al., 2012).

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Although numerous studies in Xishuangbanna rubber plantations have examined the impacts of intercropping; — focusing on soil water transport and storage, surface runoff, soil erosion, nutrient loss, plant water uptake dynamics, and interspecific water interactions within rubber-based agroforestry systems (Chen et al., 2019; Zhu et al., 2019; Yang et al., 2020) — the temporal and spatial mechanisms controlling water flow behaviors (specifically preferential and matrix flow) during infiltration across different agroforestry systems remain underexplored. Preferential flow is a ubiquitous phenomenon, and a gravity-driven process of water and solute transport through macropore-dominated pathways in soils (Bundt et al., 2001). Although universally present (Weiler and Flühler, 2004; Graham and Lin, 2011), its contribution to the total water flux varies dramatically, ranging from less than 1 % to over 90 % (Watson and Luxmoore, 1986; Leaney et al., 1993; Peterson and Wicks, 2005; Sanders et al., 2012). Preferential flow significantly influences local and regional eco-hydrological processes and biogeochemical cycles by regulating soil moisture distribution, groundwater recharge, root water uptake, solute transport, soil erosion, and related dynamics (McClain et al., 2003; Clothier et al., 2008). In contrast to preferential flow, matrix flow is primarily driven by capillary action (Germann and Beven, 1985; Zhang et al., 2017), while it also governs the amount of water stored within soil matrix. This flow is important for evaluating soil and water conservation in hillslope hydrology and watershed sedimentation (Zhang et al., 2017). The dominant mechanisms governing preferential and matrix flow are not static but dynamic, with both temporal and spatial variations. These distinct recharge mechanisms alter the spatial distribution of soil moisture and reshape subsurface flow paths (Zhao et al., 2016); consequently, influencing the dynamics of subsurface flow generation (Zhou et al., 2023). Therefore, a thorough understanding of these processes is crucial for enhancing water use efficiency, comprehensive evaluation of soil and water conservation benefits, and developing effective strategies of groundwater pollution prevention and control (Kulasekera et al., 2011).

The formation and the spatio-temporal pattern of preferential flow are regulated by the interactions among soil architecture, soil properties, landscape setting, and land use/land cover, which are, in turn, influenced by seasonal soil moisture and the characteristics of water input (Guo and Lin, 2018). In addition to these abiotic factors, root systems act as key biotic factor, exerting a significant yet various influence on preferential flow development (Li et al., 2023). On one hand, root penetration creates biopores and channels that enhance water infiltration and preferential flow (Zhou et al., 2023). On the other hand, roots may also reduce soil permeability, particularly during early growth stages when fine roots proliferate and occupy soil pores (Singh et al., 2018). Furthermore, the architecture of root systems significantly regulates infiltration pathways: well-developed horizontal roots facilitate lateral water movement in upper soil layers, while vertically oriented roots promote deeper percolation (Guo et al., 2019; Li et al., 2023). Therefore, despite the established critical influence of root systems on soil water distribution (Jacka et al., 2021; Wang et al., 2022), the mechanistic role of root biomass in water circulation remains elusive.

Various techniques have been developed and applied to characterize and visualize preferential flow in soils, including dye tracer (Zhang et al., 2022), stable isotope analysis (Wu et al., 2016), non-reactive solute breakthrough experiments (Ghafoor et al., 2013), X-ray imaging (Wildenschild et al., 2002), magnetic resonance imaging (Votrubová et al., 2003), and ground-penetrating radar (Prima et al., 2020). Among these, the dye tracer method is the most widely used for visualizing preferential flow across different spatial scales due to its high visibility, non-toxicity, minimal accumulation, and limited post-application degradation (Filipović et al., 2020; Zhang et al., 2024). Furthermore, the dye tracer technique has been adopted as a benchmark in the development of integrated approaches that provide critical insights into soil water flow dynamics (Valtera et al., 2023). For instance, dye tracer experiments combined with double-ring infiltrometers (Zhang et al.,

2017), dye infiltration experiments coupled with HYDRUS-3D modeling (Jiang et al., 2019), dye infiltration experiments used alongside soil moisture sensors (Jiang et al., 2020). However, conventional image-processing software often fails to automatically and accurately binarize dye-stained regions in cases involving uneven staining and blurred boundaries, leading to errors in the subsequent analysis of preferential flow characteristics (Bai et al., 2023). Thus, direct and quantitative assessment of preferential flow is often achieved through soil moisture sensors deployed at the catchment scale (Liu and Lin, 2015; Wiekenkamp et al., 2016). However, the use of moisture sensors alone is insufficient to objectively and comprehensively capture the spatiotemporal characteristics of soil water flow behaviors. These limitations — associated with both combined methods and standalone moisture sensors — can be partially addressed by establishing relationships between preferential flow and soil moisture dynamics, and particularly by analyzing differences in moisture content between dye-stained and non-stained regions within soil profile.

A variety of modeling approaches, including single-domain, dual-porosity, dual-permeability, and multi-porosity or multi-permeability models, have been developed to characterize the mechanisms and dynamic patterns of preferential flow (Gwo et al., 1995; Šimůnek et al., 2003). However, the effective interpretation of preferential flow through hydrological modeling remains a persistent challenge for researchers (Sternagel et al., 2019). In recent years, statistical models and ecological theories have been increasingly applied to elucidate the mechanisms underlying preferential flow, motivated by the highly nonlinear relationship between soil properties and preferential flow dynamics (Guo and Lin, 2018). For instance, species diversity theory and random forest algorithm have been successfully applied to characterize preferential flow pathways in wetland systems (Zhang et al., 2024). Nevertheless, few ecological theories have so far been introduced to address the unresolved question of how the alternating mechanisms of preferential flow and matrix flow shift across seasons. In this context, the ecological theory of bistable states offers a potential framework to explain the long-term coexistence of these two types of water flow behaviors and their transition mechanisms driven by factors such as rainfall and evaporation. Bistable states have been identified in a variety of ecosystems, including coral reefs, shallow lakes, deserts, savannas, forests, and dunes (Scheffer et al., 2001; Staver et al., 2011; Xu et al., 2020). A key feature of bistable states is the abrupt transition from one stable state to another in response to gradual environmental changes — such as variations in fire regime, rainfall, light availability, or grazing pressure (Vitense et al., 2021).

In this study, we used four rubber-related forest systems having different root biomass (decreasing in the following order: ARR > ROT > RAO > RM), based on in-situ field measurement and ecological theory of bistable states to explore the effect of the alternating mechanisms of preferential and matrix flows on soil water storage in Southwestern China. The objectives of this study are to (i) classify preferential and matrix flows of the soil vertical profile using dye tracer technique, (ii) fit preferential flow dynamics at a daily scale based on its functional relationship with soil water content (recorded by soil moisture sensors), and (iii) analyze the detailed characteristics of bistable states of preferential flow and matrix flow at daily scale in four forest systems, along with their variation in relation to root biomass. The findings are expected to enhance the understanding of spatiotemporal characteristics of soil water flow behaviors, provide a theoretical basis for refining preferential flow models in agroforestry systems, and offer a scientific foundation for selecting intercropping in rubber-based agroforestry systems in tropical regions.

2. Materials and methods

2.1. Experimental site

The experimental sites are located in the Xishuangbanna Tropical

Botanical Garden (21°55'39" N, 101°15'55" E; 594 m asl), Menglun town, Xishuangbanna prefecture, southwestern China. This region is characterized by a tropical monsoon climate, with a rainy season starting from May to October and a dry season from November to April. The monthly mean air temperature is 22.5 °C (Lin et al., 2018). The mean annual precipitation and potential evapotranspiration are approximately 1401 mm and 1389 mm, respectively (from 2011 to 2017). The soil is classified as Oxisol (USDA-NRCS et al., 2016) or Acric Ferralsols (IUSS Working Group WRB, 2015), which is derived from Cretaceous yellow sandstone (Zhou et al., 2019). The soil is approximately 2 m deep and rich in iron (Fe) and aluminum (Al) oxides due to intensive weathering and leaching (Zhao et al., 2017).

2.2. Experimental design

Four rubber-related forest systems were selected in this study (Fig. 1 A): rubber monoculture (RM), Rubber-*Alpinia oxyphylla* (RAO), Rubber-orange-tea (ROT), and artificial rubber rainforest (ARR). RAO, ROT, and ARR were rubber-based agroforestry systems. Rubber trees (*Hevea brasiliensis*) in the four systems were planted in 1990 in a deforested rainforest. Rubber trees are arranged in a double row and planted at a distance of 3 m × 4 m. Between two rubber tree rows (16 m width), *Alpinia oxyphylla* Miq. was planted in 2010 (RAO), *Citrus reticulata* and *Camellia sinensis* were planted in 2000 and 1997, respectively (ROT). The ARR site involves more than 30 species of tropical plants (e.g., *Baccaurea ramiflora* Lour., *Homalomena occulta* Lour., *Parashorea chinensis* Wang Hsie...), which were introduced from 1990 to 2005. The ARR was a multi-layer mixed species forest, and the composition of the ARR mimics the composition of native tropical rainforest species, which

was established to enhance the ecological functions of rubber forests. The four forest systems are located at similar altitudes (approximately 520 m) and gradients (approximately 5°). Except for ARR, the other forest systems were subjected to regular land management practices, such as latex tapping (approximately 120 times year⁻¹), and herbicide (applied at least yearly) and fertilizer applications.

A standard and fixed plot (10 m × 16 m) was established in each of the four forest systems, and five quadrats (1 m × 5 m) were installed in each plot (Fig. 1 B). Quadrats 1, 2 and 3 were used to measure the soil's physical properties and saturated hydraulic conductivity (K_s). Quadrat 4 was used to measure the soil volumetric water content. Quadrat 5 was used for the dye-tracer experiment to reveal the soil water infiltration pattern (Fig. 1 C). Detailed information about the stand characteristics of the four plots is shown in Table 1. The leaf area index was measured using an LAI-2200 plant canopy analyzer (Li-Cor Inc., USA). Stand litter (including leaves, flowers, fruits, and twigs) was collected using a litter trap method (three replications). Root samples (three replications) were collected by root bur in each soil layer (10 cm intervals). The particle size of soil samples (three replications) was determined by the laser granulometry method (Mastersizer E, Malvern) (Pieri et al., 2006).

2.3. Measurement of soil physical properties

Soil physical properties (e.g., soil bulk density, capillary porosity, non-capillary porosity, and total porosity) were measured on half of each quadrat 1, 2, and 3 using a cutting ring (inner diameter, 70.00 mm; height, 52.00 mm; and volume, 200 cm³) at soil depths of 10, 30, 50, 70, and 90 cm in the dry and rainy seasons. On the other half of the quadrats 1, 2, and 3, we measured soil saturated hydraulic conductivity (K_s) at the

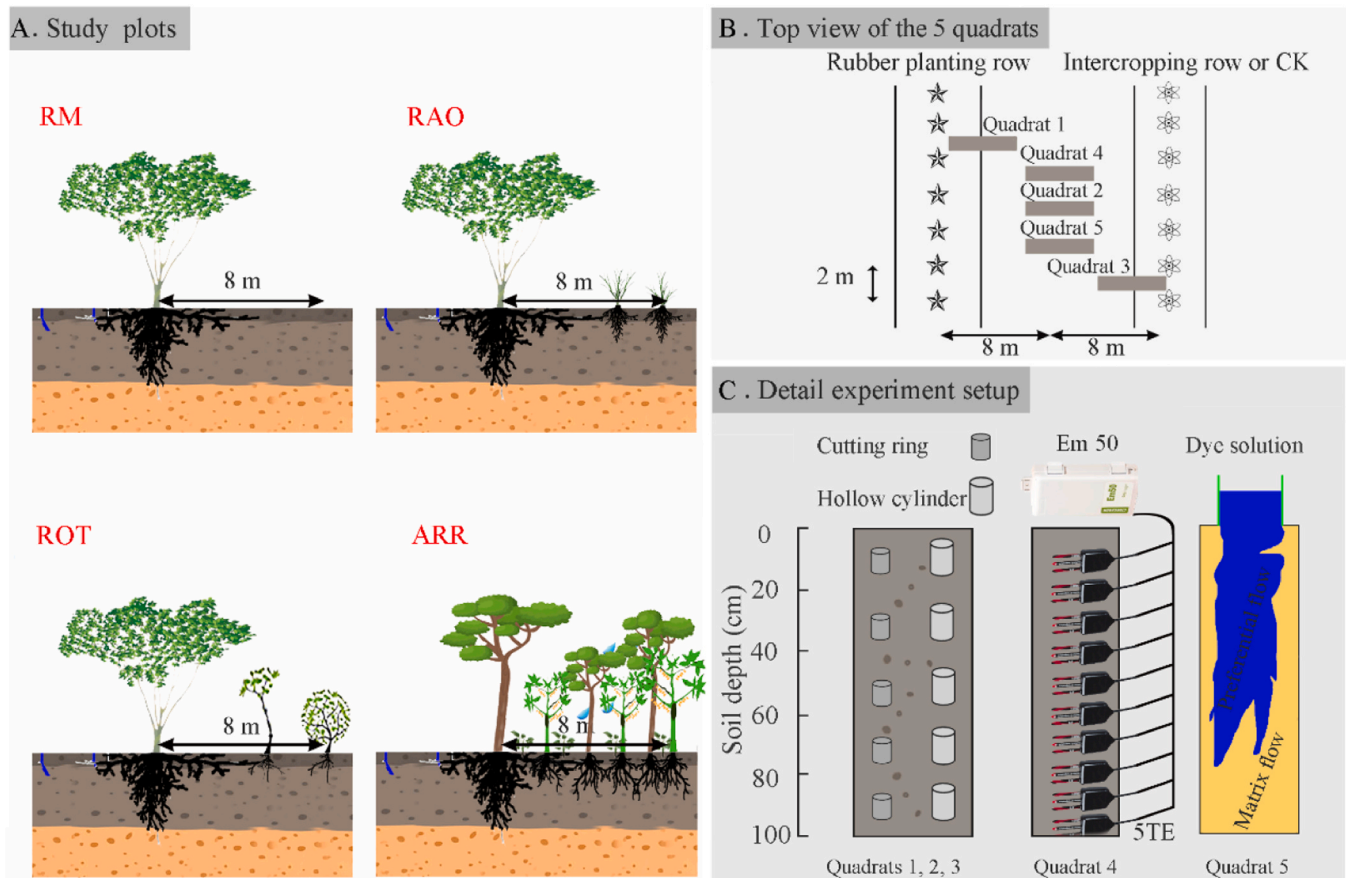


Fig. 1. Sketch maps of experimental design and measurement of field samples in the four rubber-related forest systems with different root biomass. RM, rubber monoculture; RAO, rubber-*Alpinia oxyphylla*; ROT, rubber-orange-tea; ARR, artificial rubber rainforest. The root biomass (kg m⁻²) (mean ± SD, n = 3) was 0.76 ± 0.12 in RM, 1.02 ± 0.26 in RAO, 1.39 ± 0.36 in ROT, and 1.88 ± 0.33 in ARR.

Table 1

Specific characteristics (mean $\pm$ SD, $n = 3$) of the four forest systems. RM, rubber monoculture; RAO, rubber-*Alpinia oxyphylla*; ROT, rubber-orange-tea; ARR, artificial rubber rainforest. ^a and ^b Value for whole forest systems. ^c and ^d Value for whole soil profile (0–100 cm).

Item	RM	RAO	ROT	ARR
Number of species	1	2	3	30
Leaf area index ^a (m ² m ⁻²)	1.35 $\pm$ 0.12	1.41 $\pm$ 0.21	2.21 $\pm$ 0.38	3.23 $\pm$ 0.16
Stand litter ^b (g m ⁻²)	496.58 $\pm$ 23.61	521.96 $\pm$ 27.97	539.48 $\pm$ 25.15	557.42 $\pm$ 36.42
Root biomass ^c (kg m ⁻²)	0.76 $\pm$ 0.12	1.02 $\pm$ 0.26	1.39 $\pm$ 0.36	1.88 $\pm$ 0.33
Clay ^d (%)	34.9 $\pm$ 1.0	36.4 $\pm$ 0.7	34.6 $\pm$ 1.6	34.5 $\pm$ 1.2
Silt (%)	26.0 $\pm$ 0.7	26.3 $\pm$ 0.3	26.6 $\pm$ 1.4	24.3 $\pm$ 1.2
Sand (%)	39.1 $\pm$ 0.9	37.3 $\pm$ 1.0	38.8 $\pm$ 0.5	41.2 $\pm$ 1.9
Latex tapping	Each year	Each year	Each year	No
Herbicide application	Each year	Each year	Each year	No
Mineral fertilizer	Each year	Each year	Each year	No

soil depths of 10, 30, 50, 70, and 90 cm using a single-ring infiltrometer (made up of a stainless-steel hollow cylinder) in dry and rainy seasons. Detailed information on the measurement and calculation of soil physical properties is provided by Jiang et al. (2019) and those of K_s by Zhu et al. (2019).

2.4. Measurement of soil volumetric water content

Soil volumetric water content was determined in quadrat 4 at soil depths of 10, 20, 30, 40, 50, 60, 70, 80, 90, and 100 cm using 5TE sensors (Decagon Devices, Pullman, WA, USA). The 5TE sensor data were collected at 15-minute intervals using ECH2O Utility software (Decagon Devices, Inc©, Pullman, WA) through an Em50 data logger. To avoid the impact of the sensor installation on water and soil properties, the sensor data measurement started after four months, during which several rainfall events saturated the soil. The 5TE sensor was calibrated using repacked core samples in the laboratory. Soil water storage (mm, i. e., the soil volumetric water content $\times$ soil thickness), based on soil volumetric water content, was used to express the forest systems' water conservation function (Xu et al., 2022). Moreover, a spatial distribution contour map of soil water storage was produced through ordinary kriging with the Surfer program (Version 12.0, Golden Software Inc., Golden, CO, USA) to reflect the dynamic of forest system water conservation.

2.5. Interpretation of soil water flow in vertical soil profiles

Infiltration patterns and water flow paths in quadrat 5 were interpreted using dye-tracer experiments in the dry season (Fig. 1 B and C). The surface vegetation and a thin soil layer (less than 2.0 cm) in the quadrat were carefully removed to ensure a horizontal soil surface. A four-sided metal frame of 1.2 m $\times$ 0.3 m $\times$ 0.3 m (lbh) was carefully inserted into the soil depth of 5 cm. Each metal frame was filled with Brilliant Blue FCF dye solution at a concentration of 4.0 g L⁻¹ (Flury and Wai, 2003). A filter screen (1.5 m $\times$ 0.3 m) mesh of 0.2 cm diameter was previously placed in the quadrat to ensure similar flooding conditions in the soil profiles. The dye solution was successively added to the quadrat metal frame with a constant head of 2.0 cm during the whole infiltration period. The total dye solution infiltration volume was 96 L for quadrat 5. After the disappearance of the ponding, the metal frame was covered with a plastic film to prevent evaporation and dilution by rainfall. Vertical soil profiles were dug 24 h after the end of the dye infiltration experiment. Soil profile and calibrated frame were photographed using a digital camera (Aigo H5, China) (Weiler and Flüher, 2004). The obtained images were separated into two parts: non-stained areas as matrix flow (Bundt et al., 2001) and the dye-stained areas as preferential flow

based on dye staining intensity using ERDAS IMAGINE version 9.0. The dye-stained area (cm²) of matrix flow or preferential flow was calculated for subsequent soil depths of 10 cm from the soil surface to the deep layer (100 depth). Then, the area was converted into dye area percentage (%), which is the ratio of the area of matrix flow or preferential flow to the total flow area (i.e., matrix flow + preferential flow) (Jiang et al., 2015). Detailed information on the image processing of the vertical soil sections is provided by Forrer et al. (2000), and additional information on the field measurements and image processing is given by Jiang et al. (2015).

2.6. Fitting bistable states of preferential flow and matrix flow

The infiltration pattern only provided the spatial water flow behavior (in the soil vertical profile) and did not provide information on the temporal dynamic of water flow. The shortcoming could be addressed using the spatial distribution map of soil volumetric water. First, we estimated the dye-stained area (cm²) for preferential flow in the infiltration pattern. Then, we produced a spatial distribution map of soil volumetric water on a daily scale through ordinary kriging interpolation technique with Surfer program (ver. 10, Golden software). Ordinary kriging assumes that local means are not necessarily closely related to the population mean, and therefore uses only the samples in the local neighborhood for the estimate. Detailed information on the construction of contour maps with kriging interpolation method is provided by Oliver (2010). Third, the volumetric water content (0.20 mm³ mm⁻³ or 0.24 mm³ mm⁻³ and other value) was used to divide the spatial distribution map of soil volumetric water into two parts (preferential and matrix flows), and to estimate the area (cm²) of preferential and matrix flows. Fourth, we fitted relationships in each soil layer between the area of preferential flow (assumed 10 cm² each month, with maximum value of 120 cm²) and the dye-stained area for preferential flow (the maximum value was 1000 cm²) using simple linear regressions. Fifth, we repeated the third and fourth steps to get the highest determination coefficient; and finally this analysis identified a threshold volumetric water content of 0.22 mm³ mm⁻³ (Table 2). Namely, preferential flow matched the area with volumetric water content > 0.22 mm³ mm⁻³ in the spatial distribution map of soil volumetric water. Otherwise, it was considered matrix flow.

After the spatial distribution map of soil volumetric water classification into preferential and matrix flows, the dynamic of matrix flow was fitted using a sine wave approach and plotted with SigmaPlot (version 12.5) to reflect the seasonal variation of preferential flow driven by rainfall. Thus, we obtained the bistable states of the matrix and preferential flows. Then, the levels (high or low) of preferential or matrix flow were marked differently in blue or earthy yellow based on

Table 2

Linear regressions between area for dye-stained area for preferential flow (Y) and soil volumetric water content (X) in the four forest systems under different root biomass. RM, rubber monoculture; RAO, rubber-*Alpinia oxyphylla*; ROT, rubber-orange-tea; ARR, artificial rubber rainforest.

Expression of the linear regression relation				
Area (cm ²) for volumetric water content		Dye-stained area (cm ²) for preferential flow		
> 0.22 cm ³ cm ⁻³	RM	y = 19.812x - 114.04	R ² = 0.981	P < 0.001
	RAO	y = 17.779x - 307.2	R ² = 0.890	P < 0.001
	ROT	y = 39.758x - 2403.3	R ² = 0.838	P < 0.001
	ARR	y = 25.369x - 1316.2	R ² = 0.750	P < 0.01
> 0.24 cm ³ cm ⁻³	RM	y = 22.898x - 97.836	R ² = 0.849	P < 0.05
	RAO	y = 18.361x - 172.42	R ² = 0.808	P < 0.001
	ROT	y = 17.729x - 331.97	R ² = 0.770	P < 0.01
	ARR	y = 5.6388x + 452.04	R ² = 0.071	P > 0.05
> 0.20 cm ³ cm ⁻³	RM	y = 36.85x - 1363.3	R ² = 0.766	P < 0.001
	RAO	y = 35.427x - 2275.9	R ² = 0.586	P < 0.05
	ROT	y = 33.398x - 2137.5	R ² = 0.531	P < 0.001
	ARR	y = 8.3893x + 65.718	R ² = 0.077	P > 0.05

the soil volumetric water content time series. The areas of different levels of preferential and matrix flows were summed up to compare the effects of inner function characteristics on the bistable states.

2.7. Statistical analysis

The homogeneity of variance and normality of all data was checked using Kolmogorov–Smirnov and Levene tests prior to further analysis. A two-way analysis of variance (ANOVA) was performed to evaluate the effects of the four forest systems (RM, RAO, ROT, and ARR) and the seasons on soil physical properties. Significant differences between means were detected based on the least significant difference test at $P < 0.05$. ANOVA was successfully completed to highlight the difference among the four forest systems, based on the assumption that there was no introduction of systematic error into the statistical analysis due to similar edaphic-climatic conditions at the studied sites. Linear regression was used to analyze the relationships between the area for soil volumetric water content and dye-stained area for preferential flow, then between preferential flow or soil water storage and associated factors. The analyses were performed with SPSS 20.0 (Statistical Package for the Social Sciences, USA).

3. Results

3.1. Preferential flow and matrix flow

Preferential flow and matrix flow were successfully interpreted using the dye tracer technique (Fig. 2 A). The area of preferential flow above 50 % was confined at 40 cm soil depth in RM but extended to 60 cm soil depth both in RAO and ROT and 80 cm soil depth in ARR. The total area of preferential flow in the vertical soil profile ranged in the order RM (37 %) < RAO (53 %) < ROT (56 %) < ARR (75 %), and the total area of matrix flow in the vertical soil profile ranged in the order RM (63 %) > RAO (47 %) > ROT (44 %) > ARR (25 %). As expected, matrix flow declined with increasing preferential flow. Besides, a linear relationship was found between the dye-stained area of preferential flow and the four rubber-related forest systems with different root biomass (RM to RAO, ROT, and ARR; $R^2 = 0.945$) (Fig. 3). The dye-stained area of preferential flow increased by 42 % from RM to RAO, 52 % from RM to ROT, and 103 % from RM to ARR (Fig. 3 A).

3.2. Soil water storage and soil properties

The soil water storage and soil physical properties, including soil bulk density, capillary porosity, non-capillary porosity, total porosity, and K_s , were significantly different among the four rubber-related forest systems (Table 3 and Fig. 2 B). The non-capillary porosity was significantly higher in rubber-based agroforestry systems than in RM in the rainy season (Table 3). The total porosity and K_s were significantly different among the four forest systems, and ranged in the order ARR > ROT > RAO > RM in the rainy season (Table 3). The soil water storage ranged in the order ARR > ROT > RAO > RM. The yearly soil water storage increased by 18 %, 23 %, and 31 % from RM to RAO, ROT, and ARR, respectively (Fig. 3 B). Besides, the soil water storage decreased with increasing bulk density and capillary porosity ($P < 0.05$), whereas it increased with raising non-capillary porosity, total porosity, K_s , and preferential flow ($P < 0.05$) (Fig. 4). The soil physical and hydrological properties affected the emergence of preferential flow (Fig. 5). For instance, bulk density and capillary porosity negatively affected ($P < 0.05$) the soil preferential flow, whereas the non-capillary porosity and K_s ($P < 0.05$) exhibited an opposite trend.

3.3. Alternative bistable states of preferential flow and matrix flow

Based on the significant linear regression between area for volumetric water content ($\geq 0.22 \text{ cm}^3 \text{ cm}^{-3}$) and dye-stained area for

preferential flow (Table 2), we fitted the appearance of matrix flow with the sine curve with coefficients of determination of 0.753 in RM, 0.904 in RAO, 0.916 in ROT, and 0.891 in ARR (Fig. 6). On the yearly scale, preferential flow increased while matrix flow decreased in the order RM < RAO < ROT < ARR, and the alternating occurrence of the two water flow behaviors showed a bimodal distribution pattern (Fig. 7). During the rainy season, the dominant water flow behavior was high level of matrix and low level of preferential flows in RM, while low and high levels of preferential flow dominated in ROT and ARR. During the dry season, the dominant water flow behavior was low and high level of matrix flow in RM, while high level of matrix and low level of preferential flows were more dominant in ROT and ARR. The dominant water flow behavior in RAO showed a transition from RM to rubber-based agroforestry systems. To sum up, the high level of preferential and matrix flows increased in the order of RAO < ROT < ARR both during the rainy and dry seasons.

Preferential flow changed into matrix flow when the soil volumetric water content was less than $0.22 \text{ cm}^3 \text{ cm}^{-3}$, causing an alternation of preferential flow and matrix flow over time (Fig. 8). Controlled by continuous high-intensity rainfall in the rainy season, various levels of preferential flow appeared and moved downward from surface to deep soil layers. On the contrary, under the effect of high evapotranspiration in the dry season, various levels of matrix flow occurred and flowed upward from deep to the surface soil layers. These water flow behaviors contributed to the water distribution and storage in the soil layers. To detail, during the rainy season, 13 %, 58 %, 61 %, and 72 % of water moving downward from surface to deep soil layers was controlled by the high level of preferential flow in RA, RAO, ROT, and ARR, respectively. During the dry season, 33 %, 62 %, 73 %, and 76 % of water moving upward from deep to surface soil layers was governed by the high level of matrix flow in RA, RAO, ROT, and ARR, respectively (Fig. 9). Therefore, the effect of high level of preferential and matrix flows on soil water dynamic across the soil profile increased in the order of RM < RAO < ROT < ARR both in the rainy and dry seasons (Fig. 9).

4. Discussion

4.1. Effect of rubber-based agroecosystem on soil water storage

Large-scale monoculture afforestation often reduces soil water storage and compromises regional water availability, mainly due to high root water uptake, canopy interception, surface runoff, and evapotranspiration (Dash et al., 2021). Conversely, mixed-species forests are known to enhance soil water retention (Hua et al., 2022). Our study similarly shows that RAO, ROT, and ARR systems maintained higher soil water content than monoculture (Figs. 2, 3). This soil water pattern indicates that intercropping and multi-layered canopies improved the soil water storage, corroborating the findings of previous research (Jiang et al., 2023).

The above-mentioned result was likely driven by differences in soil porosity, particularly non-capillary and total porosity, which significantly facilitated soil water storage ($P < 0.05$; Fig. 4). As shown in Table 3, both non-capillary and total porosity consistently followed the order of RM < RAO < ROT < ARR during the rainy and dry seasons. Similarly, the soil water storage exhibited the same ascending order, and this pattern was more pronounced during the rainy season. Meanwhile, the dry-season soil development — characterized by pore formation through desiccation cracking (Czarnes et al., 2000) — markedly enhanced the soil porosity. Accordingly, non-capillary porosity, total porosity, and porosity-related K_s were higher in the dry season than in the rainy season (Table 3). Nevertheless, owing to limited rainfall during the dry season, these properties exerted low influence on soil water storage. The formation of inter-aggregate cracks and voids during dry periods can generate preferential flow pathways, which become hydrologically active at the onset of the rainy season. Consequently, the increase in soil porosity and preferential flow pathways during the dry

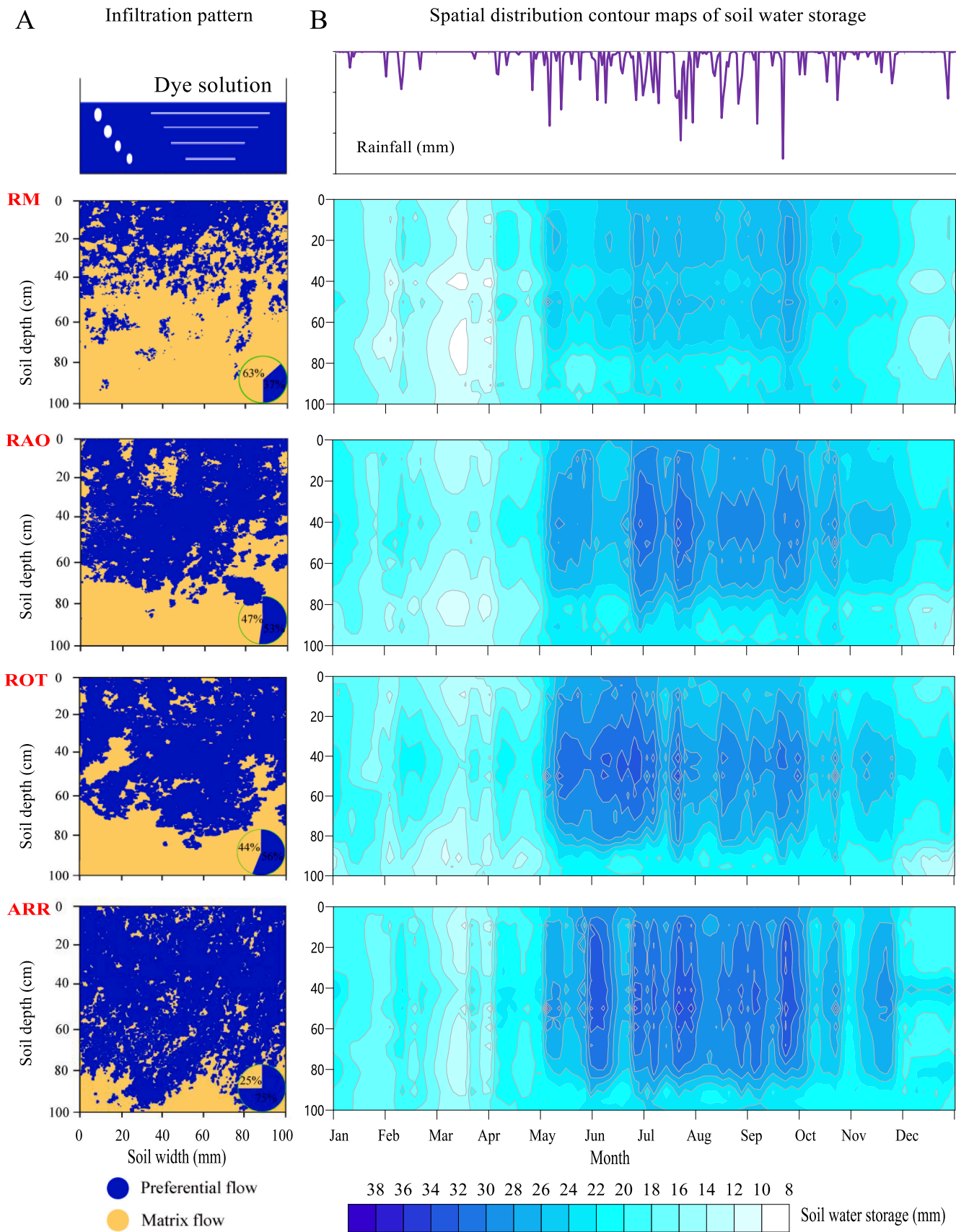


Fig. 2. Infiltration patterns at the time of measurement (A) and daily-scale spatial distribution contour maps of soil water storage (B) among the four rubber-related forest systems with different root biomass. RM, rubber monoculture; RAO, rubber-*Alpinia oxyphylla*; ROT, rubber-orange-tea; ARR, artificial rubber rainforest.

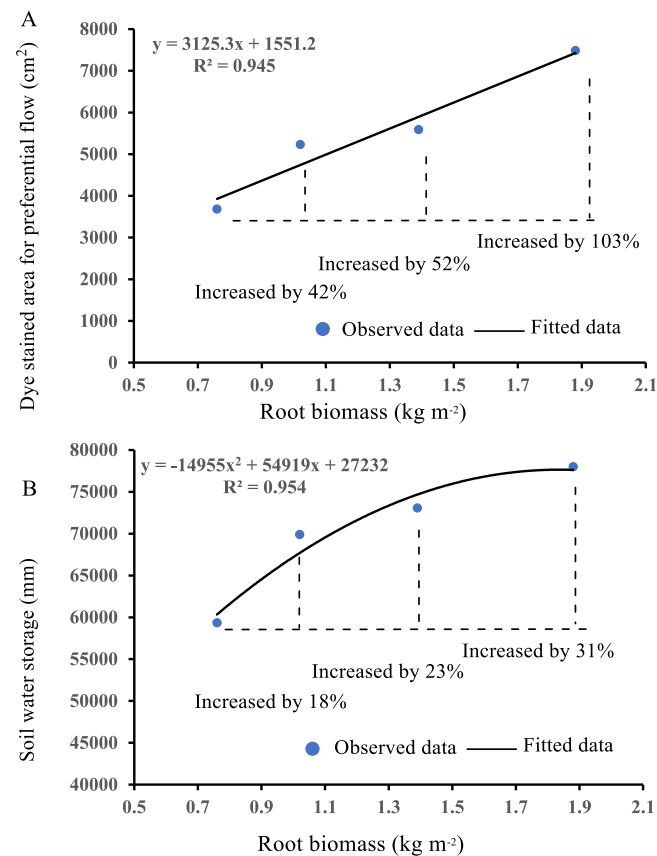


Fig. 3. Dye-stained area for preferential flow (A) and soil water storage (B) among the four forest systems with different root biomass. RM, rubber monoculture; RAO, rubber-*Alpinia oxyphylla*; ROT, rubber-orange-tea; ARR, artificial rubber rainforest. The root biomass (kg m^{-2}) (mean $\pm$ SD, $n = 3$) was 0.76 ± 0.12 in RM, 1.02 ± 0.26 in RAO, 1.39 ± 0.36 in ROT, and 1.88 ± 0.33 in ARR.

season is favorable for storing and transporting the limited initial rainfall of the subsequent rainy season (Jiang et al., 2019; Zhou et al., 2023).

Another key factor beyond soil porosity is the root characteristics (Li et al., 2023). As noted by Lin et al. (2018), the growth of rubber trees and intercroops (stems, leaves, roots) occurs primarily during the rainy season. Consequently, both living and dead root channels can create distinct pathways for water flow (Ghestem et al., 2011; Guo et al., 2019; Wu et al., 2021). As a result, the root systems and their associated structures can either enhance or inhibit preferential flow pathways (Singh et al., 2018; Zhou et al., 2023). In the present study, both preferential flow and root biomass increased in the order $\text{RM} < \text{RAO} < \text{ROT} < \text{ARR}$, and shows a strong linear relationship ($R^2 = 0.945$; Fig. 3 A). This dependance indicates that higher root biomass enhanced the formation of larger and better-connected preferential flow paths by modifying soil porosity and its connectivity, in accordance with mechanisms established in previous reports (Zhang et al., 2015; Hao et al., 2020; Jačka et al., 2021; Shi et al., 2021; Wang et al., 2022). It is important to note that, unlike the documented decline in water infiltration within vetiver grass systems beyond a root biomass threshold of 6.5 kg m^{-3} (Jotisankasa and Sirirattanachai, 2017) — which is attributed to rhizosphere compaction (Leung et al., 2015; Song et al., 2017; Lu et al., 2020) — no such reduction was observed here, as the root biomass in our agroforestry systems remained below this critical threshold.

Root biomass decomposition during the dry (non-growing) season creates a dual pore structure in the soil, forming macropores alongside numerous micropores (Guo et al., 2019; Cui et al., 2022). This structure governs water movement during low-intensity rainfall: macropores behave as preferential pathways for rapid deep infiltration, while

Table 3

Effects of seasons and forest systems on soil physical properties (mean $\pm$ SD, $n = 3$). RM, rubber monoculture; RAO, rubber-*Alpinia oxyphylla*; ROT, rubber-orange-tea; ARR, artificial rubber rainforest. K_s , soil saturated hydraulic conductivity. Note: soil physical properties are shown as mean value of 0–20, 20–40, 40–60, 60–80, and 80–100 cm depths in each system. Different letters within a season indicate a significant difference of physical properties among systems at $p < 0.05$.

Season	System	Bulk density (g cm^{-3})	Capillary porosity (%)	Non-capillary porosity (%)	Total porosity (%)	K_s (mm min^{-1})
Rainy	RM	(1.36 $\pm$ 0.01) a	(34.03 $\pm$ 0.48)a	(14.14 $\pm$ 0.29)b	(48.76 $\pm$ 0.41)d	(0.31 $\pm$ 0.03) d
	RAO	(1.32 $\pm$ 0.01) b	(32.89 $\pm$ 0.36)ab	(15.71 $\pm$ 0.50)a	(50.30 $\pm$ 0.27)c	(0.60 $\pm$ 0.03) c
	ROT	(1.26 $\pm$ 0.01) c	(32.75 $\pm$ 0.42)ab	(16.24 $\pm$ 0.28)a	(52.34 $\pm$ 0.15)b	(1.20 $\pm$ 0.07) b
	ARR	(1.20 $\pm$ 0.01) d	(31.86 $\pm$ 1.32)b	(16.40 $\pm$ 1.60)a	(54.78 $\pm$ 0.18)a	(1.45 $\pm$ 0.03) a
Dry	RM	(1.34 $\pm$ 0.02) a	(32.92 $\pm$ 0.69)a	(16.99 $\pm$ 0.24)c	(49.46 $\pm$ 0.29)c	(0.37 $\pm$ 0.03) d
	RAO	(1.29 $\pm$ 0.01) ab	(32.27 $\pm$ 0.40)ab	(17.35 $\pm$ 0.52)c	(51.50 $\pm$ 0.17)b	(0.69 $\pm$ 0.02) c
	ROT	(1.25 $\pm$ 0.01) bc	(31.50 $\pm$ 0.76)bc	(17.91 $\pm$ 0.30)b	(52.74 $\pm$ 0.19)ab	(1.28 $\pm$ 0.05) b
	ARR	(1.21 $\pm$ 0.06) c	(30.49 $\pm$ 0.71)c	(18.87 $\pm$ 0.14)a	(54.24 $\pm$ 2.35)a	(1.53 $\pm$ 0.03) a
Summary of ANOVA (p values)						
Season		0.260	< 0.05	< 0.001	0.109	< 0.05
System		< 0.001	< 0.05	< 0.001	< 0.001	< 0.001
Season $\times$ System		0.323	0.668	0.207	0.223	0.969

micropores contribute to capillary-driven water redistribution. These processes result in distinct dry-season hydrologic behaviors among the different forest systems: RM was characterized by high-low levels of matrix flow and low level of preferential flow, whereas ROT and ARR displayed a combination of high level of matrix flow and high-low levels of preferential flow. The areas of high level of matrix and preferential flow expanded with increasing root biomass ($\text{RM} < \text{RAO} < \text{ROT} < \text{ARR}$), as illustrated in Figs. 6 and 7. In other words, our findings indicate that different levels of preferential flow and matrix flow are co-existing predominant mechanisms, with their relative dominance varying by forest system. This challenges the conventional understanding, represented by other studies (Guo et al., 2019; Zhu et al., 2019; Xu et al., 2025), that preferential flow acts as the only dominant water flow behavior in such soils.

4.2. Bistable states of preferential and matrix flows mediate soil water storage

Soil water storage in monoculture and mixed afforestation has received more attention for its ecohydrological importance, along with wood production, biodiversity, soil nutrients and stress resistance (Schnabel et al., 2019; Bottero et al., 2021). The variations in ecohydrological properties between monoculture and mixed plantations should be studied due to differences in climate, soil types, and planted species (Likulunga et al., 2021). As previously discussed, the difference in soil water storage between the monoculture and mixed plantations arises from variations in root water uptake across soil depths (Wu et al., 2016) and distinct soil properties — notably macroporosity, which strongly promotes water infiltration (Zhu et al., 2019). While our prior

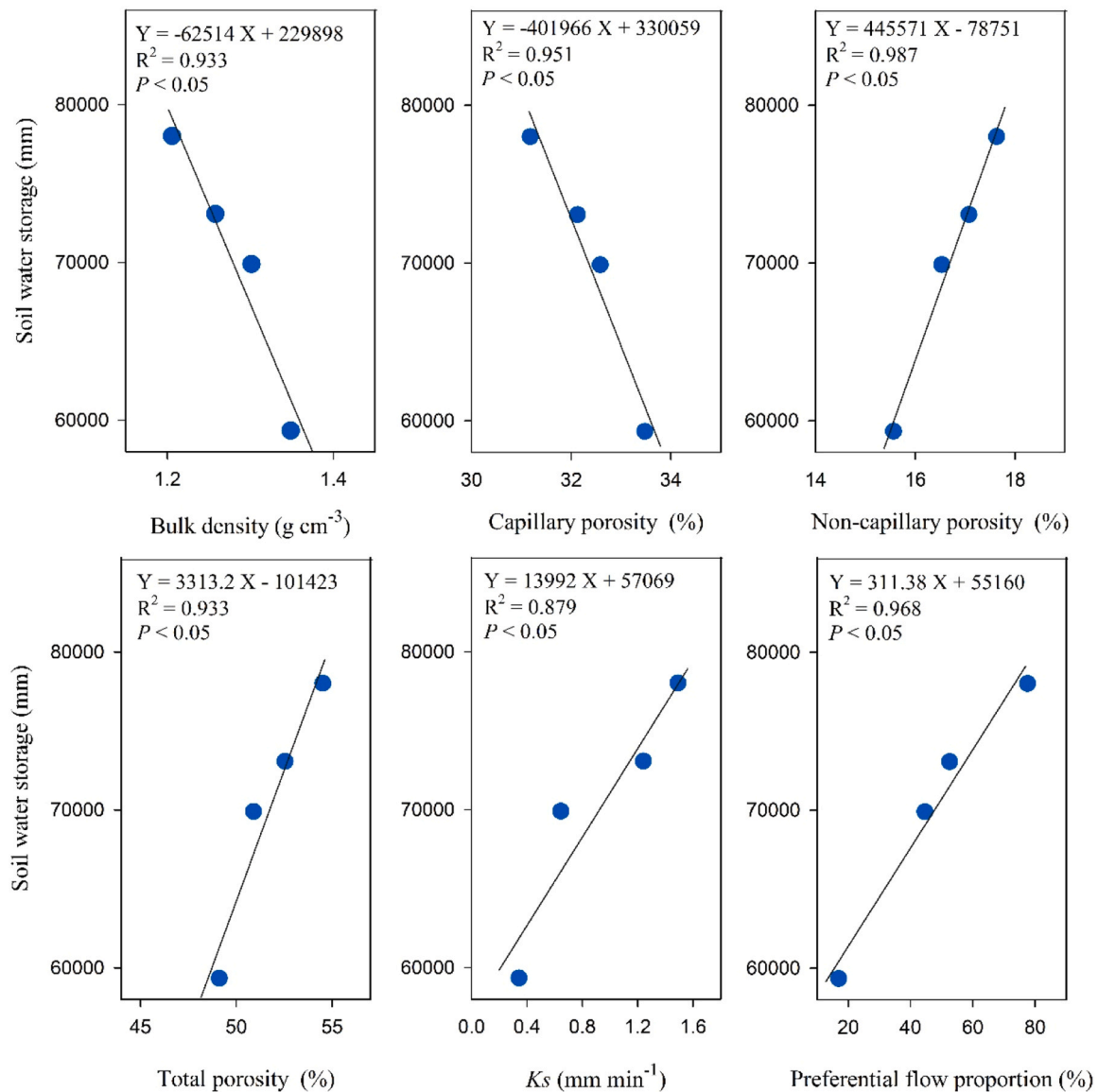


Fig. 4. Linear regression between soil water storage and associated factors. Preferential flow proportion is the ratio of preferential flow area to the total area (i.e., the sum of preferential flow area and matrix flow area).

study demonstrated that soil water flow behaviors govern the overall soil water recharge, including input from rainfall infiltration and output through deep leakage (Jiang et al., 2020), the current findings reveal a significant linear relationship between soil water storage and the extent of preferential flow (Fig. 4). This underscores that soil water storage is a complex process shaped by the dynamics of multiple soil water flow behaviors.

Soil water flow behaviors are broadly categorized into preferential and matrix flows. Science previous studies have individually examined these flows — focusing either on one type or attempting to partition preferential flow from the total flow (Zhang et al., 2017 and 2019; Filipović et al., 2020) — such approaches have nonetheless advanced our understanding of soil hydrological processes. In reality, multiple water flow behaviors often coexist simultaneously across different soils. Preferential flow, in particular, is influenced by a range of factors — such as pore characteristics, soil texture and structure, hydraulic properties, and anthropogenic impacts from forest management practices (Fig. 2 and Table 3) (Zhao and Wang, 2021). Moreover, preferential flow can occur across multiple soil layers (Xu et al., 2024), as intense rainfall events can enhance its development by connecting soil pores (Ghestem et al.,

2011). In this study, we characterized the sinusoidal pattern of matrix flow using dye tracer data and time-series soil moisture measurements, and interpreted the interplay and transition between preferential and matrix flows across soil profiles through the concept of bistable states (Figs. 6 and 7).

The analysis of bistable states revealed that the spatiotemporal characteristics of preferential flow varied across forest systems. First, within a given forest system, the occurrence of preferential flow was governed more by the persistence of rainfall than by its peak intensity or soil properties. A clear example was observed in RM; especially, preferential flow emerged under continuous rainfall of less than 20 mm min^{-1} during the rainy season (June), but failed to develop in the dry season (February) despite rainfall intensities exceeding 20 mm min^{-1} . This demonstrates that rainfall continuity, not intensity, was the decisive condition for preferential flow (Fig. 8), consistent with Xu et al. (2025) who found that rainfall duration and total amount were more influential than intensity. Second, divergent responses of preferential flow across forest systems highlighted the dominant role of soil properties. A telling example occurred in the dry season (February) under rainfall intensity exceeding 20 mm min^{-1} : preferential flow was

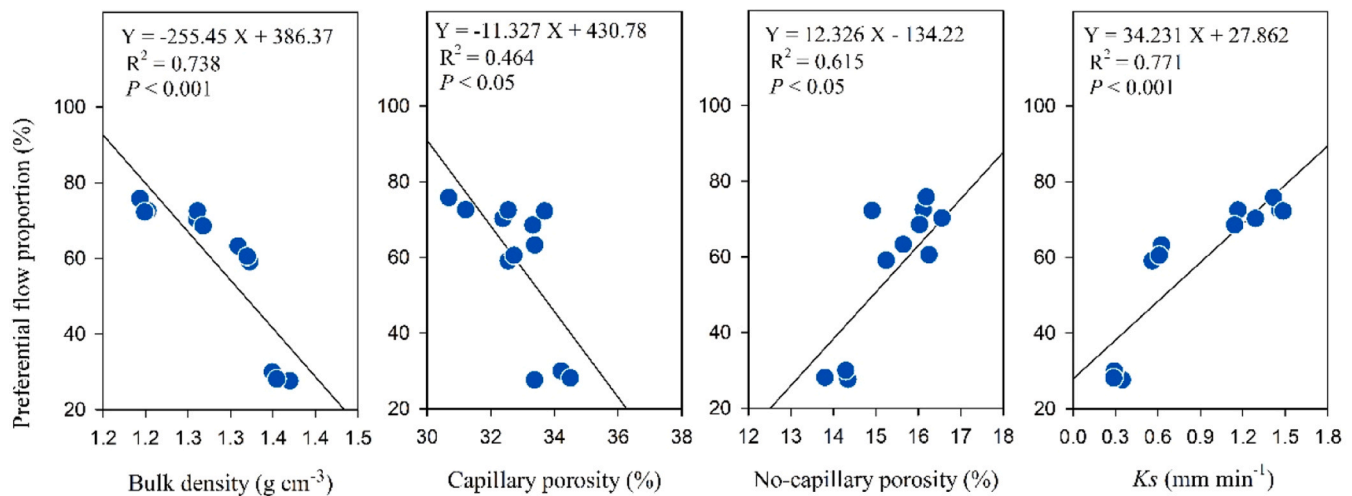


Fig. 5. Linear regression between preferential flow and soil properties during the rainy season. Preferential flow proportion is the ratio of preferential flow area to the total area (i.e., the sum of preferential flow area and matrix flow area).

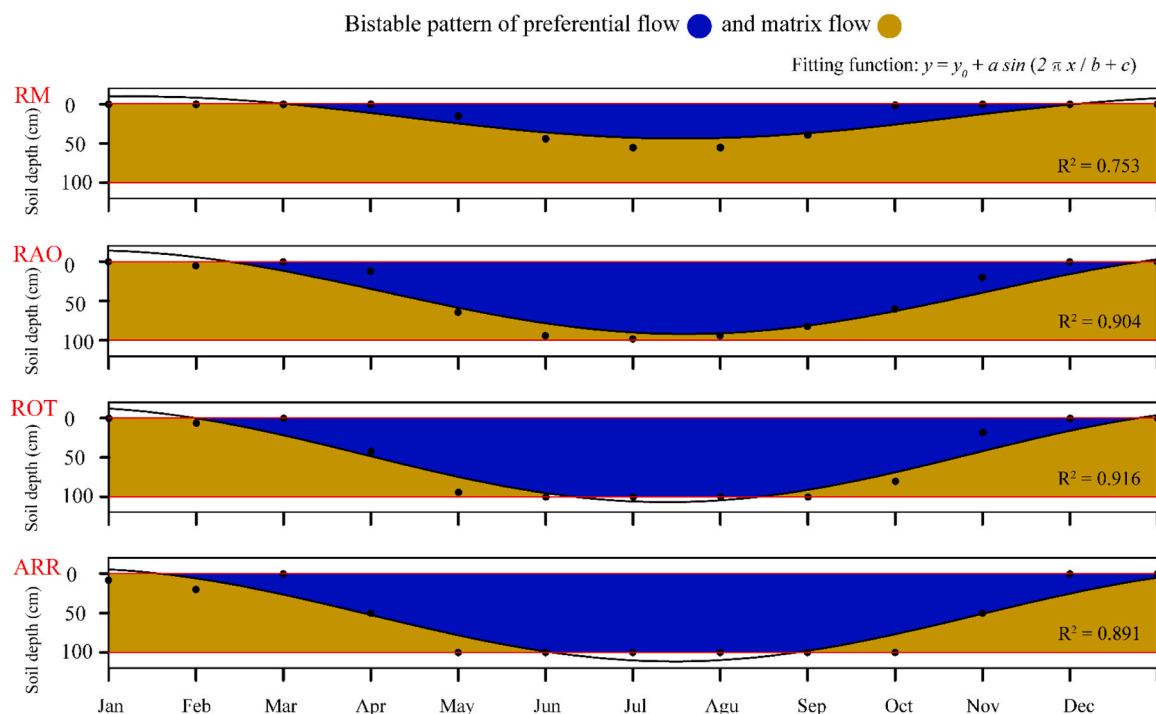


Fig. 6. Bistable states of preferential flow and matrix flow among the four forest systems with different root biomass. RM, rubber monoculture; RAO, rubber-*Alpinia oxyphylla*; ROT, rubber-orange-tea; ARR, artificial rubber rainforest.

absent in RM but present in RAO, ROT, and ARR. This disparity arose from the higher non-capillary porosity and K_s in ROT and ARR (Table 3; Fig. 5), conditions that favor preferential flow. Our results thus align with Liao et al. (2025), who also identified high non-capillary porosity and K_s as critical drivers for the initiation and enhancement of preferential flow (Fig. 5).

The proportion of preferential flow relative to total water movement exhibits considerable variability, with reported values in previous studies ranging from 1 % to over 90 % (Leaney et al., 1993; Peterson and Wicks, 2005; Sanders et al., 2012). In the present study, the observed proportions of preferential flow across different systems followed the sequence: RM (37 %) < RAO (53 %) < ROT (56 %) < ARR (75 %) (Fig. 2). Furthermore, preferential flow was categorized into high and low intensity levels (Fig. 8). The distinct spatiotemporal characteristics

associated with these two intensity classes significantly improved soil water infiltration and storage capacity during the rainy season (Figs. 6, 7, and 8). The high level of preferential flow in rubber-based agroforestry was dominant in the rainy season (especially in Jun, July, and August) when the field experienced continuous high-intensity rainfall (Fig. 8), and the preferential flow decreased from high level at the top soil surface (or rainwater concentration area) to low level at deeper soil layer (or surrounding area) (Fig. 8), leading to a radiant feature from the center to the surrounding soil. The aforementioned characteristics indicate that a substantial volume of rainwater infiltrated various soil layers and was retained in areas with high porosity, facilitated by the radial nature of preferential flow. Specifically, during the rainy season, 13 %, 58 %, 61 %, and 72 % of the downward water movement from the surface to deeper soil layers were governed by high level of preferential

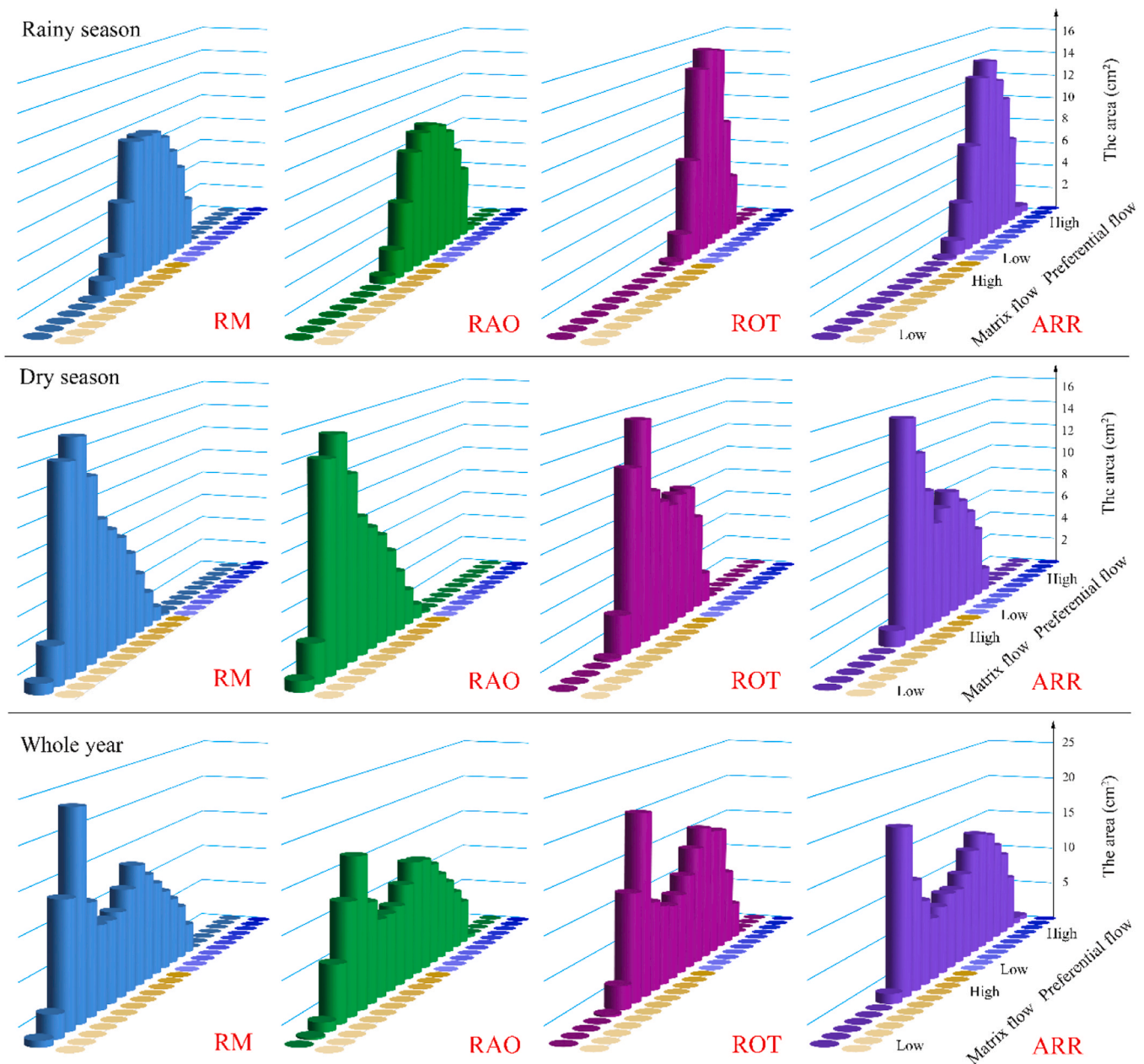


Fig. 7. Area of different levels of preferential flow and matrix flow among the four forest systems with different root biomass. Preferential and matrix flows are divided into two levels based on the volumetric water content: low [$0.22\text{--}0.26\text{ cm}^3\text{ cm}^{-3}$] and high [$0.27\text{--}0.32\text{ cm}^3\text{ cm}^{-3}$] for preferential flow; and low [$0.10\text{--}0.15\text{ cm}^3\text{ cm}^{-3}$] and high [$0.16\text{--}0.21\text{ cm}^3\text{ cm}^{-3}$] for matrix flow. RM, rubber monoculture; RAO, rubber-*Alpinia oxyphylla*; ROT, rubber-orange-tea; ARR, artificial rubber rainforest.

flow in RA, RAO, ROT, and ARR, respectively (Figs. 8 and 9). Following the cessation of preferential flow, matrix flow occurred at different levels (Fig. 8). The spatiotemporal dynamics of this matrix flow facilitated water retention within soil pores (Figs. 6, 7, and 8). More notably, during the dry season, a portion of the stored water in deeper soil layers was transported upward to surface layers via matrix flow. Especially, 33 %, 62 %, 73 %, and 76 % of the upward water movement from deep to surface soil layers during the dry season were driven by high level of matrix flow in RA, RAO, ROT, and ARR, respectively (Fig. 9). Thus, the transition from different levels of preferential flow to the levels of matrix flow was beneficial to soil water storage. Moreover, the high levels of preferential and matrix flows drove soil water flow across the soil profile, which ascended in the order of $\text{RM} < \text{RAO} < \text{ROT} < \text{ARR}$ (Fig. 9). Consequently, rainwater predominantly infiltrated to deeper layers in the rainy season, while in the dry season, water moved upward to supply

the surface soil. This seasonal reverse flow direction reveals a dynamic hydrological regime.

In summary, the bidirectional water movement — characterized by downward preferential flow from precipitation and upward matrix flow via evaporation and root uptake (Fig. 8) — enhanced soil water redistribution and storage in the rubber-related forest systems. Our study thus goes a step beyond the understanding of Zhang et al. (2019) — who demonstrated that water storage increases as rainwater infiltrates through preferential pathways and redistributes into the matrix — by revealing the complete, seasonally alternating water transport mechanism. Also, our findings underscore the need to further investigate the complex interactions among root dynamics, soil aggregation, crack formation, and matrix flow in seasonal climates. Moreover, this study suggests that intercrop density and root architecture should be carefully considered in the design of new rubber-based agroforestry systems.

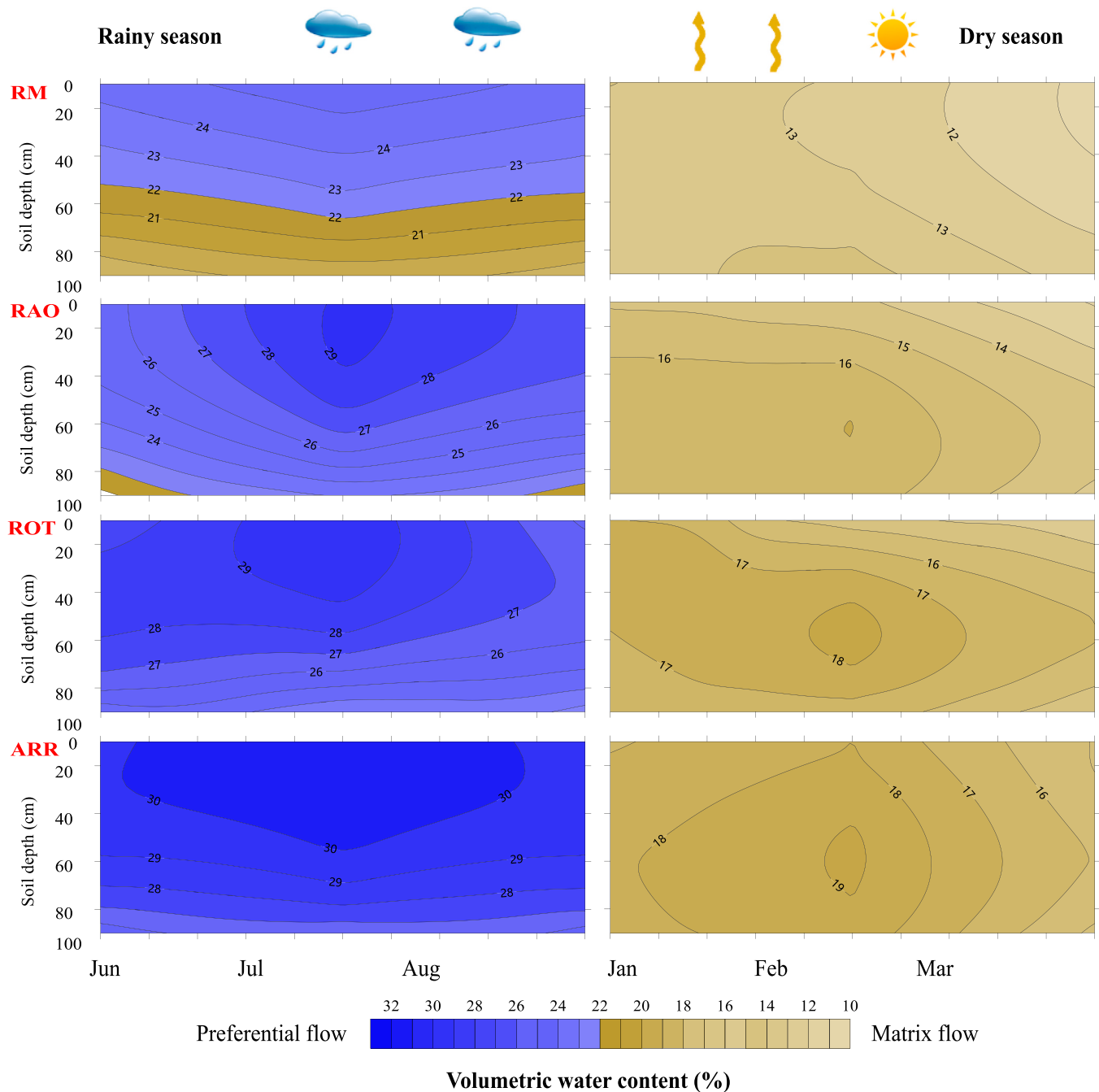


Fig. 8. Downward movement of preferential flow during the rainy season and upward movement of matrix flow during the dry season. The characteristics of preferential and matrix flows movements were different among the four forest systems with different root biomass. RM, rubber monoculture; RAO, rubber-*Alpinia oxyphylla*; ROT, rubber-orange-tea; ARR, artificial rubber rainforest.

4.3. Bistable states of soil water flow behaviors updates soil water perspectives

Several models (single-domain, dual-porosity, dual-permeability, multi-porosity or multi-permeability models) have helped to deeply understand the spatial and temporal variation of preferential flow (Gwo et al., 1995; Hutson and Wagenet, 1995; Šimůnek et al., 2003). Here, we reported information about soil water flow behaviors (preferential and matrix flows) through spatial and temporal gradients. The annual-scale bistable states between preferential and matrix flow elucidates not only the mechanisms of soil water storage but also those governing water transport through the soil profile across alternating rainy and dry seasons (Figs. 6, 7, 8, and 9). Consequently, this bistable flow framework

necessitates a re-evaluation of prevailing paradigms regarding soil water flow behavior (Fig. 10).

So far, the ecohydrological connectivity and separation, two water worlds hypothesis, and mobile-immobile water model have been developed to explain the complexity of water (and solute) transport and distribution in soil and its utilization by plants (van Genuchten and Wierenga, 1976; Xiang et al., 2019). Within the mobile water pool, preferential flow through macropores facilitates rapid water movement along the soil profile, enhancing groundwater recharge while leaving the immobile water fraction largely isolated (Seyfried and Rao, 1987). Consequently, the isotopic composition of groundwater and plant xylem water may diverge as a result of these distinct transport pathways. It follows that different soil water flow behaviors — such as preferential

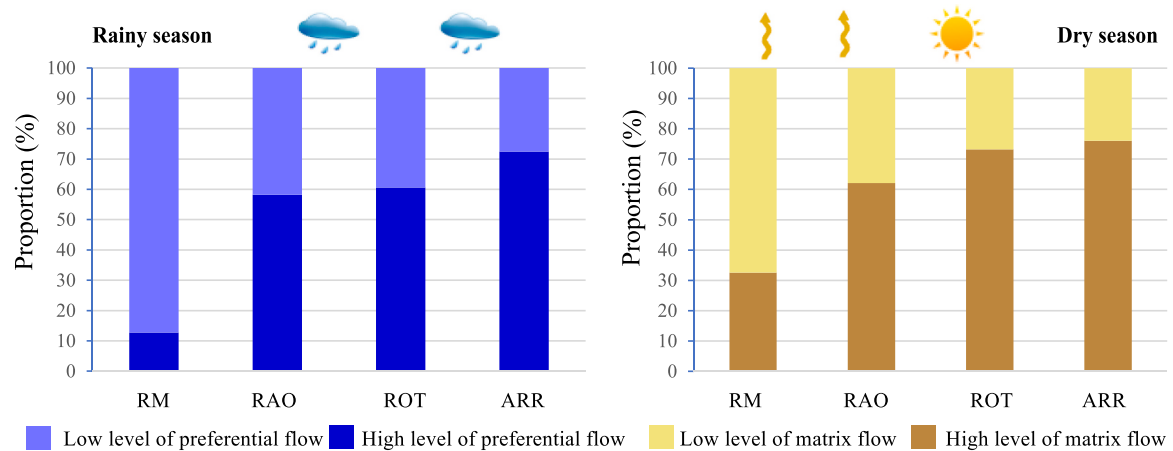


Fig. 9. Effects of various levels of preferential and matrix flows on water changing across soil profiles during the rainy and dry seasons. RM, rubber monoculture; RAO, rubber-*Alpinia oxyphylla*; ROT, rubber-orange-tea; ARR, artificial rubber rainforest.

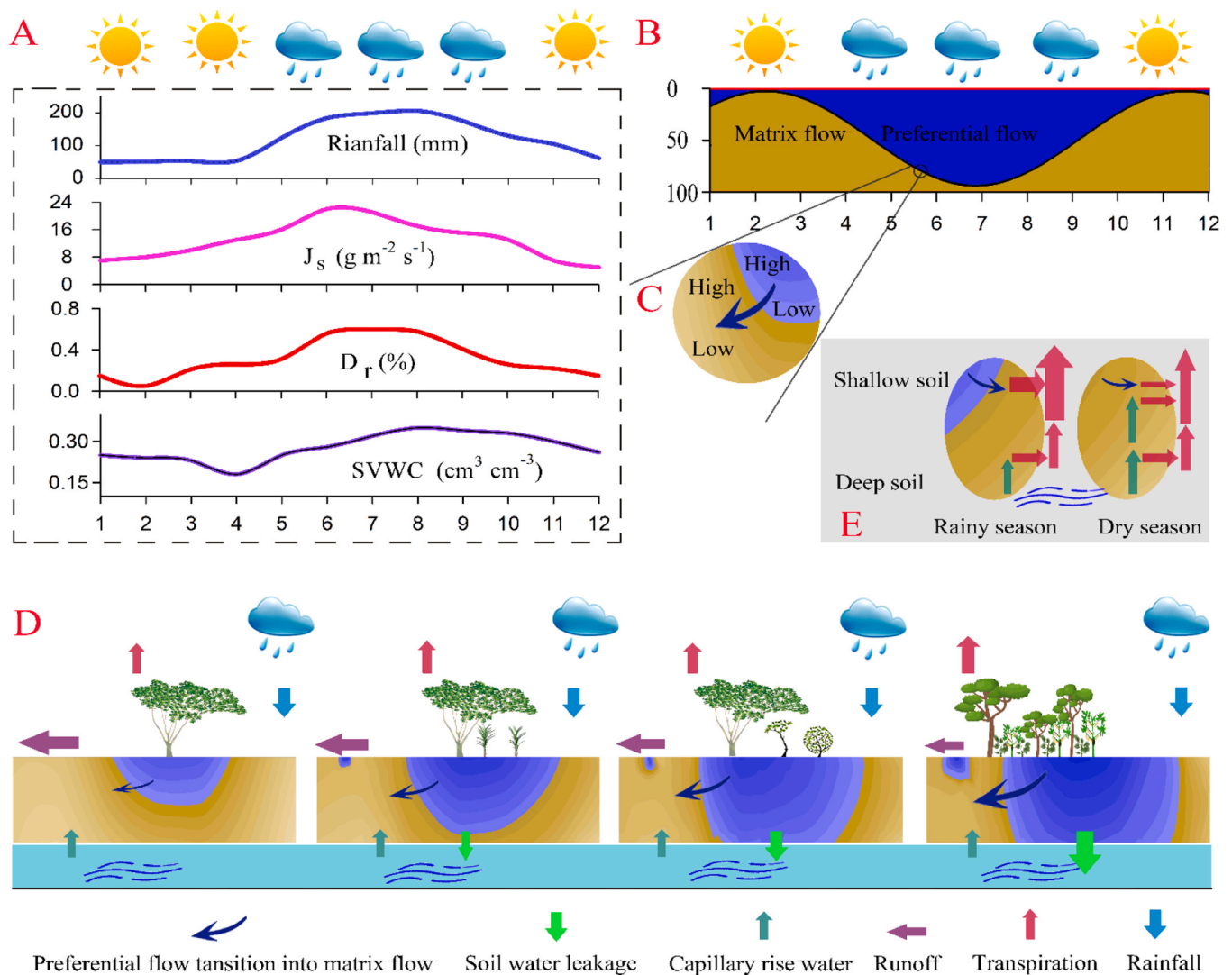


Fig. 10. Conceptual model for the bistable states of soil water flow behaviors. A: Current research progress in soil water related to rainfall features in rubber forestry (Lin et al., 2018). B: Bistable states of preferential flow and matrix flow in the four forest systems. C: Transformation of preferential flow into matrix flow. D: Bidirectional upward and downward water flows across soil profiles of the four forest systems. E: Different water sources for transpiration. RM, rubber monoculture; RAO, rubber-*Alpinia oxyphylla*; ROT, rubber-orange-tea; ARR, artificial rubber rainforest. The root biomass followed the order $\text{ARR} > \text{ROT} > \text{RAO} > \text{RM}$. SVWC, soil volumetric water content. D_r : monthly growth rate of the diameter at breast height, J_s : sap flow density on a per-area-of-trunk basis.

flow, matrix flow, wetting front velocity, and convection-diffusion rates — can influence solute concentrations and isotopic signatures in both layered soil water and groundwater. If further research confirms these observations, it could provide critical insights for optimizing sampling strategies and interpretation of isotopic data. Additionally, it may offer supporting evidence for the debated ‘two water worlds’ hypothesis, which proposes a hydrological separation between plant-accessible water and mobile groundwater (Evaristo et al., 2015; Hervé-Fernández et al., 2016; Qian et al., 2017).

The bistable states of preferential and matrix flows (Fig. 10 B) also raise potential questions regarding the use of stable isotope techniques (δD and $\delta^{18}O$) to study plant water sources (Wu et al., 2016; Yang et al., 2020). Previous studies have shown that plants mainly absorb water from upper soil layers recharged by rainfall during the rainy season, and shift to deeper soil layers recharged by groundwater during the dry season. The bistable states of preferential and matrix flows, however, suggest that soil water constitutes a continuous system, where water can be exchanged between upper and deeper soil layers through gravitational drainage and capillary rise (Fig. 10 C, D, and E). When water in the upper soil is depleted, water from deeper layers can move upward to replenish it. Therefore, it can be speculated that when upper soil water is depleted to the permanent wilting point, plants in the dry season will utilize water from the upper soil that comprises two sources: the residual water originally stored in the upper layer, and the water supplied from deeper soil layers via upward movement (Fig. 10 E). Future studies should test this hypothesis and quantify the contributions of these two sources by integrating multiple methods and theoretical frameworks, such as dye tracer techniques, stable isotopes, sap flow and lysimeter measurements, water balance theory, and the concept of bistable states of soil water flow behaviors.

5. Conclusions

This study employs an integrated methodology — including dye tracing, soil moisture time-series analysis, and the sine wave approach — to assess the effects of rubber-related forests with different root biomass on the bistable states of soil water flow behaviors. Our analysis reveals distinct soil water dynamics and hydrological drivers across four tropical forest systems (with root biomass ranking: $ARR > ROT > RAO > RM$), particularly regarding the mechanisms governing water flow and storage under alternating rainy and dry seasons.

The results demonstrate that forest systems significantly modulate soil water storage (following the order $RM < RAO < ROT < ARR$), a pattern governed by the bistable states of preferential and matrix flow and their seasonal alternation. Preferential flow exhibited a radial feature, facilitating the rapid diffusion and storage of water into the soil matrix. The decline in the low level of preferential flow coincided with the rise of high level of matrix flow, reflecting a transition process that enhances water distribution and storage across shallow and deep soil layers during the rainy season. Meanwhile, during the dry season, matrix flow originating from deeper layers supplied water upward to the shallow soil. In summary, the bistable states between preferential and matrix flows are critical to forest ecosystem stability and development, as they enhance soil water storage capacity and maintain a dynamic equilibrium of bidirectional water movement throughout the soil profile.

Our findings deliver formal field evidence for moisture-driven alternative water flow behaviors in rubber-related forest soils, thereby offering a new perspective for local water management and address the limitations of previous studies that have considered sole preferential flow to study water management measures of rubber-based agroforestry systems only from a static perspective. The results also give new perspective for deep understanding of the dynamic characteristic of soil water across soil profile during alternating rainy and dry seasons. Finally, we suggest the use of the theory of multiple stable states to explain transitions among co-occurring water flow behaviors

(preferential flow, matrix flow, lateral flow, and so on) under multiple hydrological processes (rainfall, fog, evaporation, transpiration, runoff, leakage, and so on) in Xishuangbanna, thus opening a new theoretical pathway for future ecohydrological research.

CRedit authorship contribution statement

Xiao-Jin Jiang: Writing – review & editing, Writing – original draft, Visualization, Investigation, Conceptualization, Project administration, Funding acquisition. **Guan-Yuan Ma:** Investigation, Methodology, Writing – original draft. **Zakari Sissou:** Writing – review & editing, Writing – original draft, Visualization. **Ashutosh Kumar Singh:** Writing – review & editing, Visualization. **Haofei Wang:** Investigation, Methodology, Writing – original draft. **Shi-Yi Wen:** Investigation, Methodology, Writing – original draft. **Xiai Zhu:** Investigation, Writing – original draft. **Xin Zou:** Investigation, Methodology. **Zi-Qiang Yuan:** Writing – review & editing, Writing – original draft, Supervision. **Youxing Lin:** Writing – review & editing, Writing – original draft, Investigation, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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